The Simple pendulum

Aim of the Experiment:

Determination of the acceleration of gravity by using a simple pendulum.

Apparatus:

A simple pendulum – a stop watch – a metric ruler.

Theory:

The simple pendulum consists of a mass, m, suspended vertically by a non-elastic light thread of length (l).

This simple pendulum is an important physical apparatus, since it provides a simple and easy way for determination of the gravitational acceleration (g) accurately.

By using the polar coordinates which its origin is at the suspension Point as shown in the fig (1) we have:

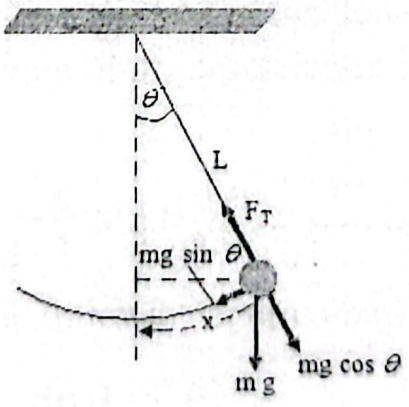


Figure (1)

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Newton's law of motion of the pendulum is:

$$\vec{F} = m\vec{g}$$

$$m\vec{a} = m\vec{g} \quad \ldots\ldots\ldots\ldots (1)$$

Since m is the mass of the bob and $\vec{a}, \vec{g}$ are the acceleration of the motion and of the gravity receptively.

One resolve equation (1) into two components as follows

$$m\left[l\ddot{\theta}\hat{\theta} + (\ddot{l} - l\dot{\theta}^2)\hat{r}\right] = (-g\sin\theta\,\hat{\theta} + mg\cos\theta\,\hat{r})$$

Since:

$$l = constant \qquad \therefore \ddot{l} = 0$$

$$\dot{\theta} = \frac{d\theta}{dt}, \ddot{\theta} = \frac{d^2\theta}{dt^2}$$

From equation (1) we have

$$l\ddot{\theta} = -g\sin\theta \quad \ldots\ldots\ldots (2)$$

$$-l\dot{\theta}^2 = g\cos\theta \quad \ldots\ldots\ldots\ldots (3)$$

Equation (3) does not give us the tension force in the threat, this is not important for determination of the acceleration of gravity, but equation (2) can be written as

$$\ddot{\theta} = -\frac{g}{l}\sin\theta \quad \ldots\ldots\ldots\ldots\ldots (4)$$

if θ is small enough , the following approximation can be applied

$$\sin\theta \cong \theta \quad \ldots\ldots\ldots\ldots (5)$$

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and equation (4) becomes:

$$\ddot{\theta} = -\frac{g}{l}\theta \ldots\ldots\ldots\ldots\ldots (6)$$

This is an equation of simple harmonic motion of a periodical oscillation time τ, which defines as the time required for the bob to undertake one complete oscillation. By solving (6), it is found that the periodical time τ is given by

$$\tau = 2\pi\sqrt{\frac{l}{g}} \ldots\ldots\ldots\ldots (7)$$

$$\therefore \quad \tau^2 = \frac{4\pi^2}{g} l \quad \ldots\ldots\ldots (8)$$

By using equation (8), an experiment for determination of the acceleration of gravity (g) can be done. Since

$$g = 4\pi^2\left(\frac{l}{\tau^2}\right) \ldots\ldots\ldots\ldots\ldots\ldots\ldots (9)$$

Procedure:

1- Set up the apparatus as shown in the figure (2)

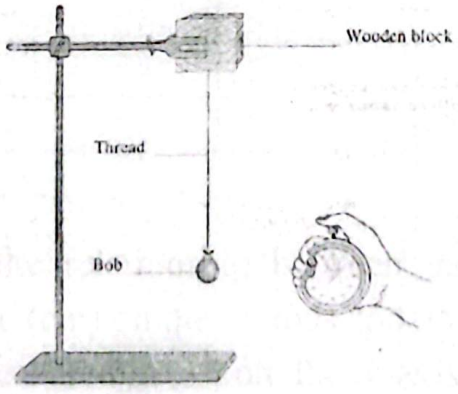


Figure (2)

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2-Measure the length of the Pendulum from the Point of Suspension to the ball center.

3-Slide the ball through a small angle and let pendulum oscillates in a vertical plane. Starts register the time of 20 complete oscillations by using a suitable accurate stop-watch. let it be T' seconds .

4-Calculate the periodical time (τ) of one oscillation as follows

$$\tau = \frac{T'}{20} \ldots\ldots\ldots\ldots\ldots (10).$$

4- Repeat steps from 2 to 4 for different lengths such as 80cm, 70cm, 60cm, 50,cm and 40cm.

5-Write down your data in a table as follows.

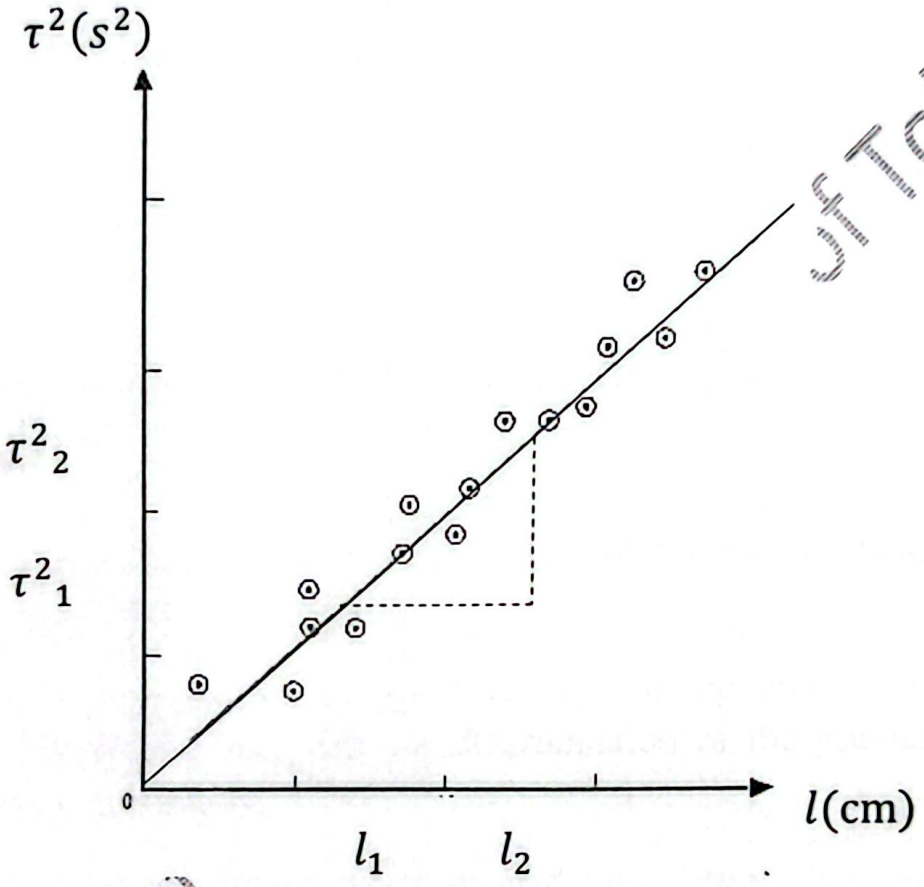
Length of the thread (L) cm	Time taken for (20) Oscillations in s	Average Period of one Oscillation(τ) in s	τ^2 in s^2
20			
30			
40			
50			

6- Draw graphically the relationship between the length of the thread (l) measured in (cm) on the x-axis and the square of the periodical time (τ^2) measured in (s^2) on the y-axis. The graph obtained should be a straight line satisfies equation (9)

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7- Calculate the slope of the obtained straight line by selecting two points as shown as follows.



$$The\ slope = \frac{\tau_2^2 - \tau_1^2}{l_2 - l_1} = \frac{\Delta\tau^2}{\Delta l} \quad \ldots\ldots\ldots (11)$$

8- By using equation (9), calculate the gravitational acceleration of the gravity at the location of the experiment as follows:

$$g = \frac{4\pi^2}{the\ slope} \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (12)$$

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